

6: Bayesian decision analysis and practical significance with ROPE, emmeans, and marginaleffects

Bayesian Mixed Effects Models with brms for Linguists

Job Schepens

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1 The Problem: Does Effect X Matter?

1.1 Research Scenario

Imagine you’ve run a psycholinguistic experiment and fitted a Bayesian model. You have posterior distributions for your effects. Now you face the question:

“Is this effect meaningful in practice?”

- You have an estimate (e.g., $\beta = 0.12$ log-RT units)
- You have uncertainty (95% CI: [0.08, 0.16])
- But: Is this difference big enough to matter?

Practical significance is different from **statistical significance**:

- Statistical: “Is there an effect?” (distinguishable from noise) (we will discuss this in the session on Bayes Factor)
- Practical: “Does it matter?” (large enough to care about)

1.2 Why Practical Significance Matters

Consider these scenarios:

Scenario A: Statistically credible but trivial

- Effect: 2ms difference in reading time
- 95% CI: [1ms, 3ms] (clearly non-zero!)
- But: Readers can’t perceive < 20 ms differences
- **Conclusion:** Real effect, but irrelevant

Scenario B: Uncertain but possibly large

- Effect: 50ms difference (noticeable to readers)
- 95% CI: [-10ms, 110ms] (wide uncertainty)
- Traditional conclusion: “not significant”
- But: Most credible values are meaningful
- **Conclusion:** Worth investigating further

2 Where We Are in the Analysis Workflow

2.1 The Bayesian Workflow So Far

Let's review where we've been:

1. **Module 01:** Set priors
2. **Module 02:** Prior predictive checks
3. **Module 03:** Posterior predictive checks
4. **Module 04:** Compare priors → Sensitivity analysis
5. **Module 05:** Compare models → Evaluate predictive performance (LOO)
6. **Today (Module 06):** Practical significance (ROPE + comparisons)
7. **Module 07:** Bayes factors & hypothesis comparison
8. **Module 08:** Convergence

2.2 When to Use Each Approach

QUESTION	TOOL
Which model structure is better?	LOO (Module 05)
Is effect meaningful?	ROPE (Module 06)
Compare multiple groups?	emmeans (Module 06)
Custom predictions/contrasts?	marginaleffects (M06)
Evidence for hypothesis?	Bayes Factor (M07)
Are estimates robust to priors?	Prior comparison (M04)

2.2.1 Decision Framework

Use **ROPE** when:

- You want to declare “effect too small to matter”
- You need clear decision rules (accept/reject/undecided)

Use **emmeans** when:

- You have factorial designs (multiple groups/conditions)

Use **marginaleffects** when:

- You're working with complex models (GAMs, interactions)

Use **Bayes Factors (Module 07)** when:

- You want to quantify evidence for one hypothesis over another (see Module 07)

Use **LOO** (Module 05) when:

- Comparing different model structures
- Doing feature selection
- Predictive performance is primary concern

2.3 Setup

2.4 Generate Reaction Time Data

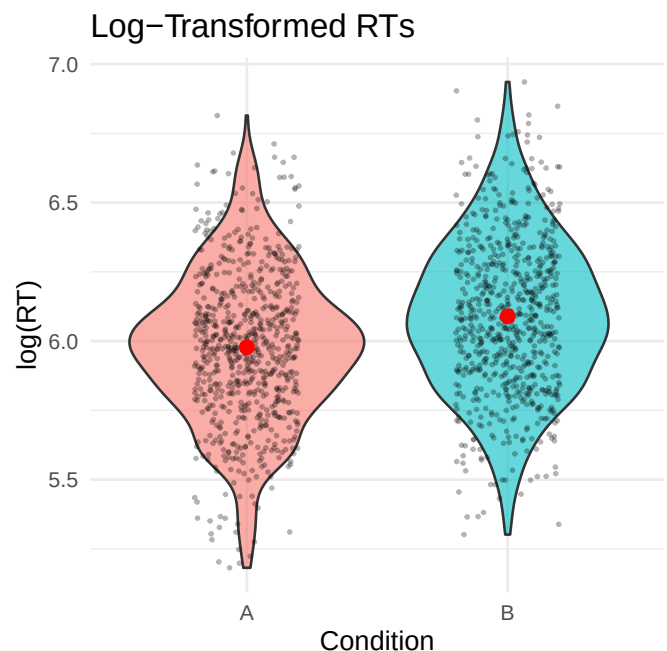
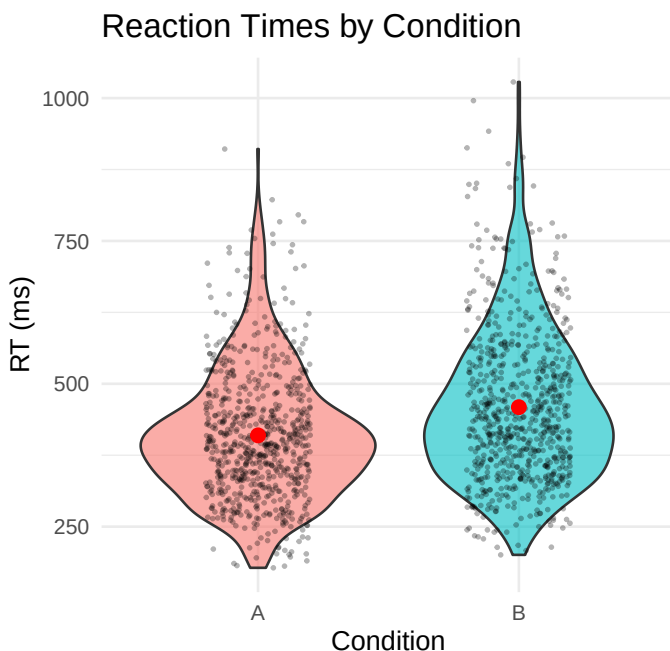
We'll generate RT data similar to previous modules, but with specific properties useful for hypothesis testing demonstrations:

- Clear directional effect (Condition B slower than A)
- Effect size in a realistic range for psycholinguistics
- Adequate sample size for stable estimates

```
# A tibble: 2 x 6
  condition     n mean_rt median_rt sd_rt mean_log_rt
  <fct>      <int>   <dbl>    <dbl> <dbl>    <dbl>
1 A         720    410      395   114     5.98
2 B         720    459      439   133     6.09

# A tibble: 1 x 2
  Measure          Value
  <chr>          <dbl>
1 Effect size (log scale) 0.111
```

2.5 Visualize the Data



2.6 Fit the Model

2.7 Model Summary

```
Family: gaussian
Links: mu = identity
Formula: log_rt ~ condition + (1 + condition | subject) + (1 | item)
Data: rt_data (Number of observations: 1440)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000

Multilevel Hyperparameters:
~item (Number of levels: 24)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.11     0.02   0.08   0.16 1.00     870    1458

~subject (Number of levels: 30)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.17     0.02   0.13   0.22 1.00     790
sd(conditionB)    0.06     0.02   0.02   0.09 1.00     879
cor(Intercept,conditionB) 0.03   0.26  -0.45   0.54 1.00    2095
Tail_ESS
sd(Intercept)    1296
sd(conditionB)    726
cor(Intercept,conditionB) 2035

Regression Coefficients:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    5.98     0.04   5.90   6.05 1.01     478     992
conditionB    0.11     0.02   0.08   0.14 1.00    2917    2927

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    0.20     0.00   0.19   0.21 1.00    4011    3063

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

3 Region of Practical Equivalence (ROPE)

3.1 Theory: What is “Practically Equivalent”?

Statistical significance tells us if an effect differs from zero. But:

- **Statistical question:** Is the effect *exactly* zero?
- **Practical question:** Is the effect *close enough* to zero to ignore?

This is where **ROPE** (Region of Practical Equivalence) comes in.

3.1.1 The ROPE Framework (Kruschke, 2011)

Instead of testing if $\theta = 0$ exactly, we define a small interval around zero:

$$\text{ROPE} = [-\varepsilon, +\varepsilon]$$

where ε is the **smallest effect size we care about** (domain-specific).

Decision rules:

1. **95% HDI entirely inside ROPE** $\rightarrow$ Accept practical equivalence (effect negligible)
2. **95% HDI entirely outside ROPE** $\rightarrow$ Reject equivalence (effect matters)
3. **95% HDI overlaps ROPE** $\rightarrow$ Uncertain (collect more data or accept uncertainty)

3.1.2 Setting ROPE Boundaries

! Common Pitfall: Post-Hoc ROPE Boundaries

ROPE boundaries must be set **before seeing results** based on:

- Domain knowledge (e.g., “RT differences $< 50\text{ms}$ are imperceptible”)
- Standardized effect sizes (e.g., “Cohen’s $d < 0.1$ considered negligible”)

Mistake: Setting ROPE after looking at posterior to get desired conclusion.

Why it matters: Post-hoc boundaries invalidate the test (like p-hacking).

3.1.3 Four Methods for Justifying ROPE Boundaries

From Kruschke (2015, Chapter 12): ROPE boundaries should represent the smallest effect you care about. Here are four principled methods for setting them:

3.1.3.1 Method 1: Previous Research / Meta-Analysis

Use effect sizes from prior literature to calibrate what counts as “small.”

Method	Typical Effect	ROPE Boundaries	Interpretation
Based on Meta-Analysis	0.10 log-units (100ms or 10%)	$[-0.03, 0.03]$	Effects $< 3\%$ are small relative to field norms

When to use:

- Mature research area with existing effect size estimates
- You want to compare to “typical” effects in the field
- You have access to meta-analyses or large-scale studies

Example:

In their meta-analysis of 50 reading time studies, Smith et al. (2020) report a mean effect of $d = 0.35$ for syntactic complexity manipulations.

We set our ROPE at $d = 0.10$ (approximately 1/3 of the typical effect), corresponding to ± 0.03 log-RT units.

3.1.3.2 Method 2: Measurement Precision

ROPE should exceed measurement error—otherwise you’re testing noise.

Method	Measurement Error	ROPE Boundaries	Interpretation
Based on Measurement Precision	0.02 log-units (20ms)	$[-0.03, 0.03]$	Effects smaller than measurement error are unreliable

When to use:

- You have reliability/measurement error estimates
- You want to avoid claiming effects smaller than noise
- Measurement precision is the limiting factor

How to estimate measurement error:

```
# A tibble: 2 x 2
  Measure      Value
  <chr>      <chr>
1 Within-subject SD 0.085
2 Suggested ROPE   ±0.127
```

3.1.3.3 Method 3: Perceptual/Practical Thresholds

Effects below perception or practical impact are negligible by definition.

Method	Perceptual Threshold	ROPE Boundaries	Interpretation
Based on Perceptual Threshold	50ms (0.05 log-units)	$[-0.05, 0.05]$	Effects imperceptible to readers are negligible

When to use: - You can measure perceptual/practical thresholds - You care about real-world impact (not just statistical detection) - You have pilot data or existing benchmarks

Example applications: - **Reading:** Just-noticeable difference in reading time - **Accuracy:** Minimal detectable accuracy change - **Rating scales:** Smallest perceived difference on Likert scale - **Medical:** Minimum clinically meaningful difference (MCID)

How to measure perceptual thresholds:

Pilot study design:

1. Show participants trials from both conditions
2. Ask: "Did you notice any difference in difficulty/speed?"
3. Correlate subjective reports with actual RT differences

4. Find threshold below which participants don't notice
→ Use this threshold as ROPE

3.1.3.4 Method 4: Standardized Effect Sizes (Cohen's d)

Use conventional effect size benchmarks from your field.

Method	Cohen's d Threshold	Estimated Pooled SD	ROPE Boundaries	Interpretation
Based on Stan- dardized Effect Size	0.1 (half of 'small' effect)	0.2 log-units	[-0.02, 0.02]	Effects with $d < 0.10$ are 'very small'

When to use: - No domain-specific benchmarks available - You want to follow field conventions - Standardized metrics are expected in your area

Common benchmarks: - **Cohen's d:** Small = 0.2, Medium = 0.5, Large = 0.8 - **Correlation (r):** Small = 0.1, Medium = 0.3, Large = 0.5

- **R²:** Small = 0.01, Medium = 0.09, Large = 0.25

Note: These are **conventions**, not laws of nature! Domain-specific thresholds (Methods 1-3) are usually better.

3.1.4 For RT Data (log-scale)

For our analysis:

- **ROPE** = [-0.05, +0.05] on the log scale
- **On original scale:** This corresponds to RT ratios between 0.95 and 1.05
 - Since we model $\log(\text{RT})$, a difference of ± 0.05 on the log scale = $e^{\pm 0.05}$ = multiplying RT by 0.95 to 1.05
 - Example: If Condition A = 400ms, ROPE means effects between 380ms and 420ms ($\pm 5\%$)
- **Interpretation:** RT differences smaller than 5% are too small to be practically meaningful

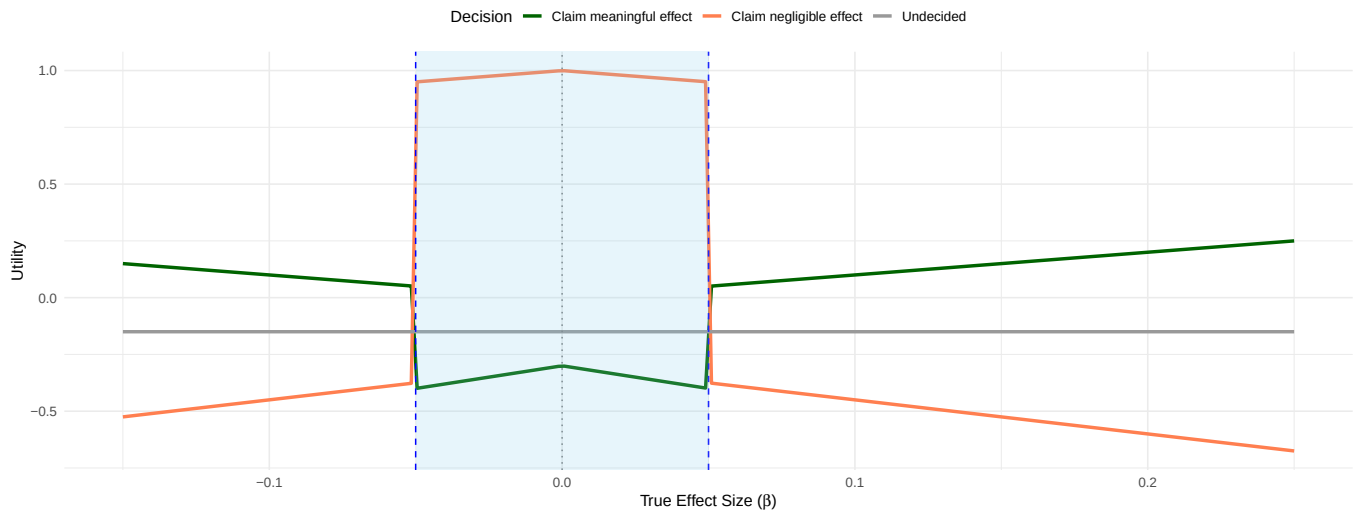
3.1.5 The Implicit Utility Function

Let's make explicit what ROPE boundaries mean in decision-theoretic terms (Gelman et al., 2013, Chapter 9; [Stan User's Guide on Decision Analysis](#)):

Scenario: You must decide whether to claim "Condition B is meaningfully slower" or "Condition B is essentially equivalent to A".

Utility function (simplified for illustration):

Utility Function Across Effect Sizes
ROPE boundaries $[-0.05, 0.05]$ define where utilities cross



What this shows:

1. **Inside ROPE** ($|| < 0.05$): “Claim negligible” has highest utility
2. **Outside ROPE** ($|| > 0.05$): “Claim meaningful” has highest utility
3. **At ROPE boundaries** ($|| = 0.05$): “Undecided” becomes optimal
 - At $= \pm 0.05$ exactly:
 - claim_meaningful utility: $-2(0.05) - 0.3 = \mathbf{-0.40}$ (wrong decision, heavy penalty)
 - claim_negligible utility: $-1.5(0.05) - 0.3 = \mathbf{-0.375}$ (wrong decision, moderate penalty)
 - undecided utility: $\mathbf{-0.15}$ (cost of more data)
 - The grey horizontal line (-0.15) is higher than both colored lines at the boundaries
 - Both wrong decisions incur the -0.3 penalty plus proportional loss, making them worse than collecting more data
4. **Asymmetric losses**: False positive penalty stronger than false negative
 - **Green line inside ROPE**: Drops to $-2 \times || - 0.3$ (steep slope, heavy penalty)
 - **Coral line outside ROPE**: Drops to $-1.5 \times || - 0.3$ (moderate slope, lighter penalty)
 - This asymmetry means: claiming a meaningful effect when it’s negligible (false positive) is penalized more heavily
5. **The ROPE boundaries (± 0.05) are where the utility functions cross zero**
 - Inside: coral (claim negligible) is positive, green (claim meaningful) is negative
 - Outside: green (claim meaningful) is positive, coral (claim negligible) is negative
 - This defines the decision boundary
6. **Why lines have slopes**: Utility depends on effect size magnitude
 - Larger true effects $\rightarrow$ higher utility for correct “meaningful” claim (green line increases)
 - Effects closer to zero $\rightarrow$ higher utility for correct “negligible” claim (coral increases toward 1 as $\rightarrow 0$)
 - Being wrong by a lot is worse than being wrong by a little
7. **Why “Undecided” is optimal near boundaries**:
 - The -0.3 constant penalty on wrong decisions ensures that near $|| = 0.05$, both claim decisions have utility < -0.15
 - In practice: When posterior uncertainty is high and overlaps ROPE boundaries, the expected utility of “undecided” often wins

- The grey line represents the cost of reducing uncertainty through more data collection
- The grey line (cost of more data = -0.15) becomes the least-bad option
- In practice: This happens when posterior uncertainty is high and overlaps the ROPE boundaries

! ROPE Boundaries = Utility Crossover Points

Your ROPE boundaries should be set where the utilities of “claim meaningful” and “claim negligible” are equal. This is your **smallest effect size of interest (SESOI)**.

Computing expected utilities:

	decision	expected_utility
1	claim_meaningful	0.1114245
2	undecided	-0.1500000
3	claim_negligible	-0.4671367

3.1.6 Interpretation

The optimal decision is to **claim_meaningful** (maximizes expected utility).

i From Decision Theory to ROPE

The traditional ROPE decision rule: - “If 95% HDI excludes ROPE → Reject null”
 ...is actually an approximation to: - “Choose the decision with maximum expected utility”

The HDI-based rule works well when:

1. Losses are approximately symmetric
2. We want to control error rates at ~5%
3. We prefer simple rules over computing expected utilities

For asymmetric losses or complex decisions, computing expected utilities explicitly (as shown above) provides more principled decisions.

3.2 Understanding ROPE: Three Decision Outcomes

3.2.1 What ROPE Really Is (Formal Definition)

From Kruschke (2015, p. 336):

A region of practical equivalence (ROPE) indicates a small range of parameter values that are considered to be **practically equivalent** to the null value for purposes of the particular application.

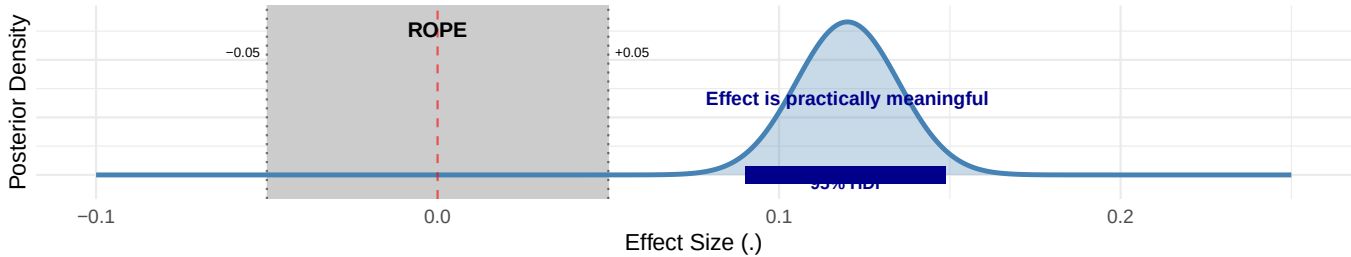
Note: ROPE analysis has **three possible outcomes**, not just two!

Three Possible ROPE Outcomes

ROPE boundaries: $[-0.05, +0.05]$ | Shaded gray region = ROPE | Thick blue line = 95% HDI

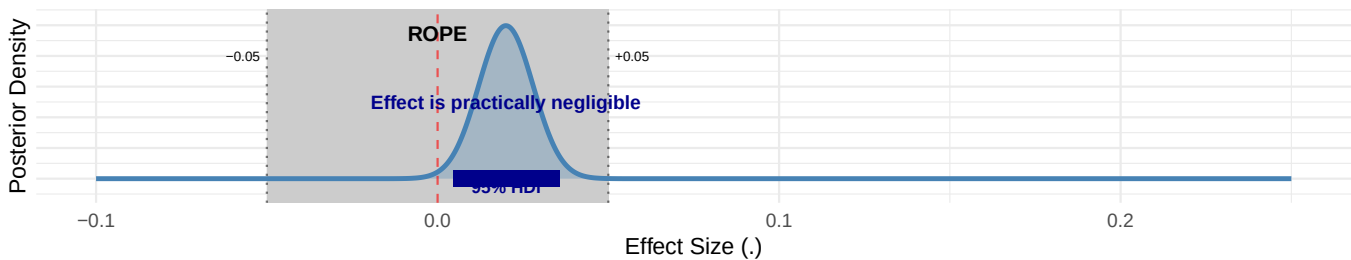
Decisive: Reject H.

HDI excludes ROPE



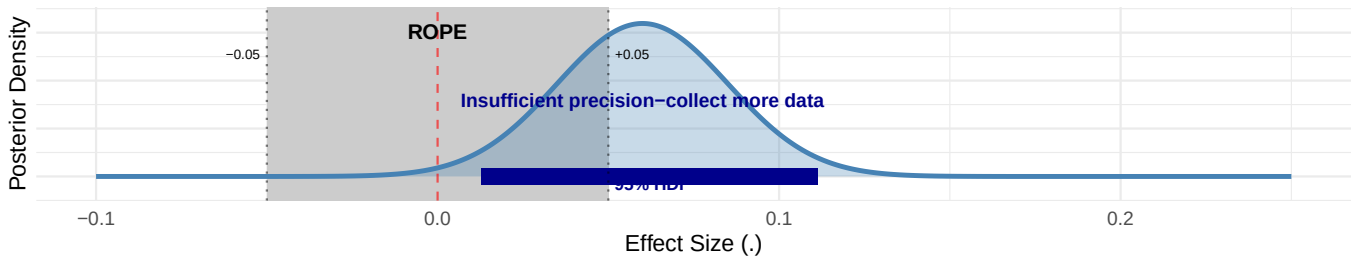
Decisive: Accept H.

HDI inside ROPE



Undecided

HDI overlaps ROPE



! Three Outcomes, Not Two!

Current practice shows:

1. **HDI excludes ROPE** → Reject H (effect is meaningful)
 - Example: 95% HDI = $[0.09, 0.15]$, ROPE = $[-0.05, 0.05]$
 - Interpretation: “Effect is decisively larger than our practical significance threshold”
2. **HDI inside ROPE** → Accept H (effect is negligible)
 - Example: 95% HDI = $[0.01, 0.03]$, ROPE = $[-0.05, 0.05]$
 - Interpretation: “Effect is decisively smaller than our practical significance threshold”
3. **HDI overlaps ROPE** → UNDECIDED (insufficient precision)
 - Example: 95% HDI = $[0.03, 0.09]$, ROPE = $[-0.05, 0.05]$
 - Interpretation: “We cannot make a clear decision—some credible values are meaningful, others aren’t”
 - **This is not a failure!** It’s honest reporting of uncertainty

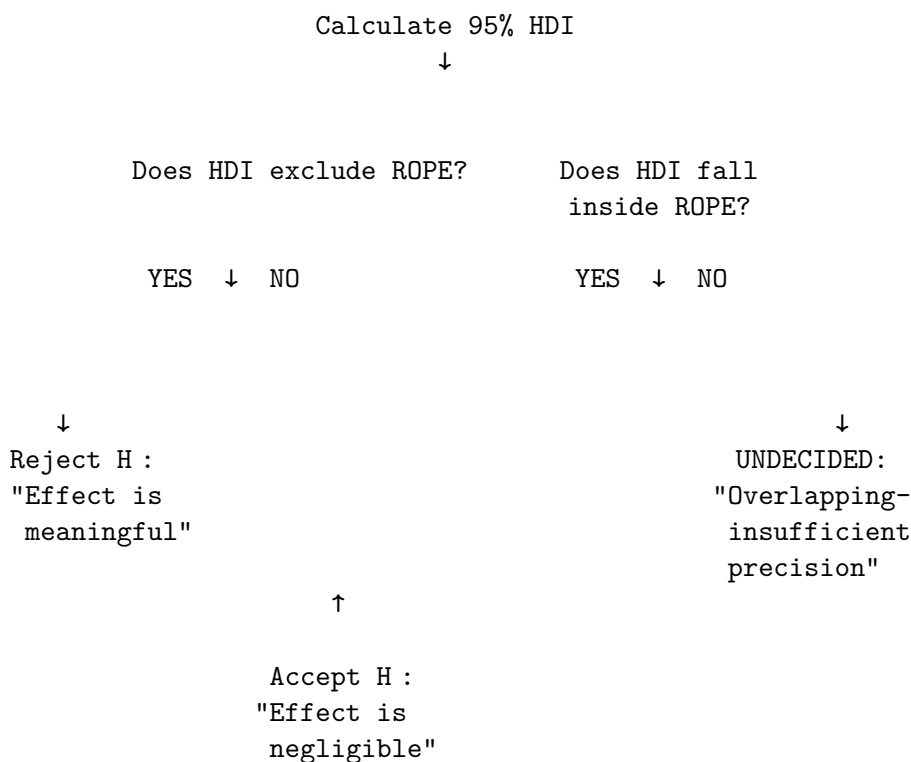
i “Undecided” Is a Feature, Not a Bug

From Kruschke (2015, p. 338):

“Be clear that any discrete decision about rejecting or accepting a null value does not exhaustively capture our knowledge about the parameter value. Our knowledge about the parameter value is described by the **full posterior distribution**.”

When HDI overlaps ROPE: - You have learned something: The effect might or might not be meaningful - Your options: 1. Collect more data to narrow the posterior 2. Accept the uncertainty and make a practical decision based on other factors 3. Use a less conservative threshold (e.g., 89% HDI instead of 95%) - **Don’t** force a binary decision when the data don’t support one!

3.2.2 Decision Flowchart



i Manual ROPE Calculation (HDI-Based Approximation)

ROPE is simpler than computing utilities explicitly and works well for most research questions.

In practice, we usually use the HDI-based ROPE approximation rather than computing expected utilities explicitly. This is computationally simpler and works well when: - Losses are approximately symmetric (false positive false negative) - We want standard 95% decision threshold - We prefer simple rules over custom utility functions

Now let’s use the traditional ROPE decision rules (which approximate the decision-theoretic framework):

```
# A tibble: 5 x 2
```

Measure	Value
<chr>	<chr>

```

1 ROPE boundaries [-0.05, 0.05]
2 95% HDI        [0.084, 0.142]
3 % Below ROPE   0.0%
4 % Inside ROPE  0.0%
5 % Above ROPE   100.0%

```

[1] ****REJECT equivalence****: Effect is practically meaningful (positive) - 95% HDI entirely above

Concepts So Far

Before moving on, you should understand: - ROPE boundaries represent the smallest effect you care about - Three possible decisions: Accept H₀, Accept H_a, or remain undecided - Decision based on whether 95% HDI overlaps, excludes, or is inside ROPE - This approximates choosing the decision with maximum expected utility

Decision Rule = Maximizing Expected Utility

What just happened in decision-theoretic terms:

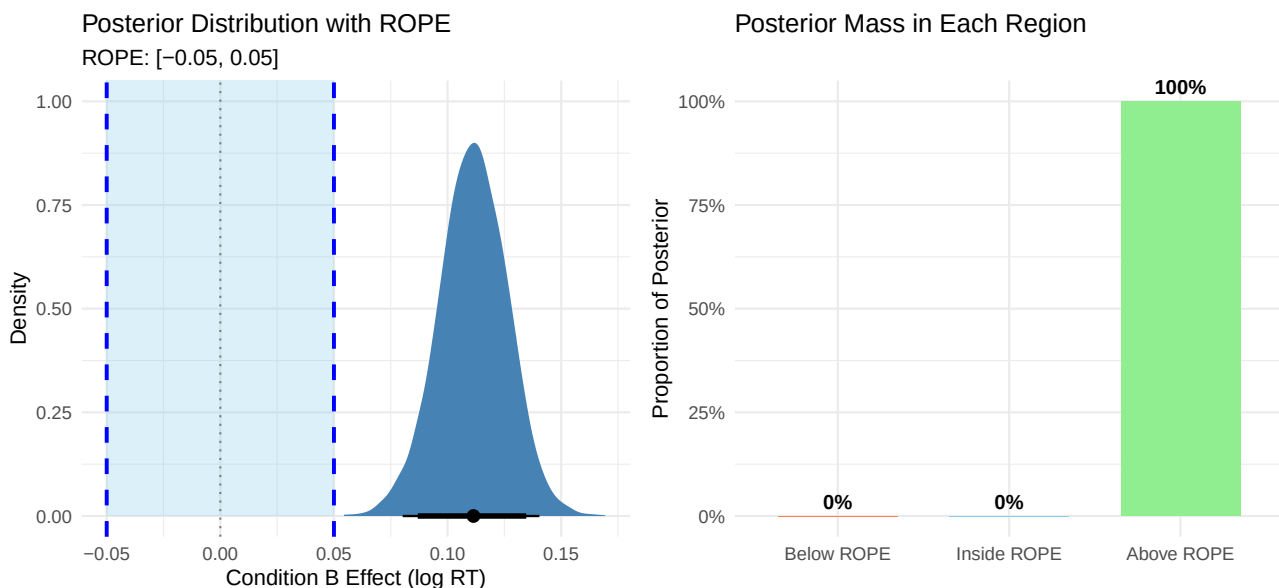
1. **We computed posterior probabilities**: $P(\text{in each region} \mid \text{Data})$
2. **We applied a decision rule**: Based on where 95% HDI falls
3. **This approximates**: Choosing decision with maximum expected utility

The connection: - When 95% HDI > ROPE: $E[U(\text{"meaningful"})] > E[U(\text{"negligible"})]$ with high confidence - When 95% HDI < ROPE: $E[U(\text{"negligible"})] > E[U(\text{"meaningful"})]$ with high confidence - When overlapping: Expected utilities too close to call → “undecided” optimal

The 95% threshold encodes a loss function where: - We’re willing to accept 5% risk of wrong decision - Losses are approximately symmetric for false positives vs. false negatives - If your losses are **asymmetric** (e.g., false positives much worse), you should: - Use stricter threshold (e.g., 99% HDI), OR - Compute expected utilities explicitly (as shown in previous section)

3.2.3 Visualizing ROPE

Manual ROPE Analysis



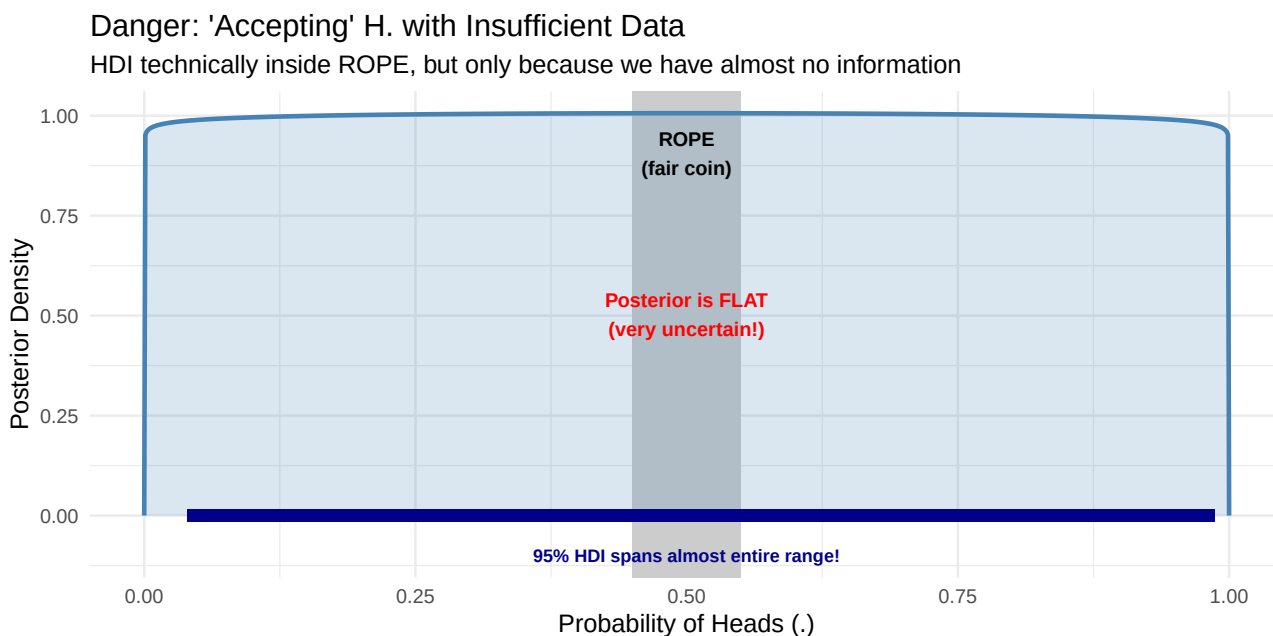
3.3 Warning: ROPE Requires Adequate Precision!

⚠ The Precision Problem

From Kruschke (2015, Section 12.2.1.1): Even with very little data, you can “accept” the null if the posterior is flat enough!

This is one of the most overlooked caveats about ROPE analysis.

i Dangerous Example: Accepting H_0 with Poor Data



```
# A tibble: 3 x 2
  Measure      Value
  <chr>        <chr>
1 Data        1 head out of 2 flips
2 95% HDI for [0.039, 0.986]
3 HDI Width    0.947
```

Kruschke’s solution (p. 348):

“High precision demands a large sample size... But when we are trying to accept a specific value of , it seems logically appropriate that we should have a reasonably precise estimate indicating that specific value.”

3.3.1 Before Using ROPE to Accept H_0 : Check These Three Things

```
# A tibble: 8 x 2
  Check      Value
  <chr>      <chr>
1 "1. HDI Width" 0.0576
2 "  Status"    FAIL (> 0.05) - Need more data
3 "2. Effective Sample Size (bulk)" 2387
4 "  Effective Sample Size (tail)" 2584
```

```

5 "    Status"                                PASS (> 1000)
6 "3. Posterior SD"                          0.0146
7 "    SD/ROPE Ratio"                        0.15
8 "    Status"                                PASS (< half ROPE)

```

3.3.2 Scenario Examples

Scenario 1: Wide HDI Inside ROPE - 95% HDI: [0.01, 0.04] (width = 0.03) - ROPE: [-0.05, 0.05] - WRONG: “Effect is negligible” - RIGHT: “HDI inside ROPE, but width (0.03) suggests moderate uncertainty. More data needed.”

Scenario 2: Narrow HDI Inside ROPE - 95% HDI: [0.015, 0.025] (width = 0.01) - ROPE: [-0.05, 0.05] - RIGHT: “Effect is negligible. HDI is narrow (0.01) and falls well within ROPE.”

Scenario 3: HDI Overlaps ROPE Boundary - 95% HDI: [0.03, 0.08] - ROPE: [-0.05, 0.05] - RIGHT: “Undecided. Some credible values fall within ROPE, others exceed it. More data needed.”

! Rules of Thumb for Precision

Before using ROPE to **accept H** (claim negligible effect), verify:

For RT effects (log-scale): - HDI width < 0.05 log-units - Posterior SD < 0.025 (half the typical ROPE width) - *Why?* Larger SD means too much uncertainty about whether the effect is truly negligible.

For accuracy (probability scale): - HDI width < 0.10 - Posterior SD < 0.05

For all analyses: - ESS (bulk) > 1000 - ESS (tail) > 1000 - Convergence diagnostics passed (R < 1.01)

If these checks fail: - Don’t claim effect is negligible - Report “insufficient precision” or “undecided” - Consider collecting more data

💡 When You Can Trust ROPE to Accept H

Strong evidence for negligible effect requires:

1. **Narrow posterior:** HDI width < half ROPE width
2. **Central location:** HDI midpoint near zero
3. **Good sampling:** ESS > 1000, R < 1.01
4. **Model fit:** Posterior predictive checks pass

Example of strong evidence:

Effect: = 0.02 (95% HDI: [0.01, 0.03])

ROPE: [-0.05, 0.05]

HDI width: 0.02 (narrow)

ESS: 2500 (adequate)

→ Can confidently claim negligible effect

Example of weak evidence:

Effect: = 0.02 (95% HDI: [-0.01, 0.05])

ROPE: [-0.05, 0.05]

HDI width: 0.06 (too wide)

ESS: 800 (marginal)

→ Cannot confidently claim negligible effect

3.4 Using bayestestR Package

The `bayestestR` package provides convenient functions for ROPE analysis.

```
# Proportion of samples inside the ROPE [-0.05, 0.05]:
```

Parameter	Inside ROPE
Intercept	0.00 %
conditionB	0.00 %

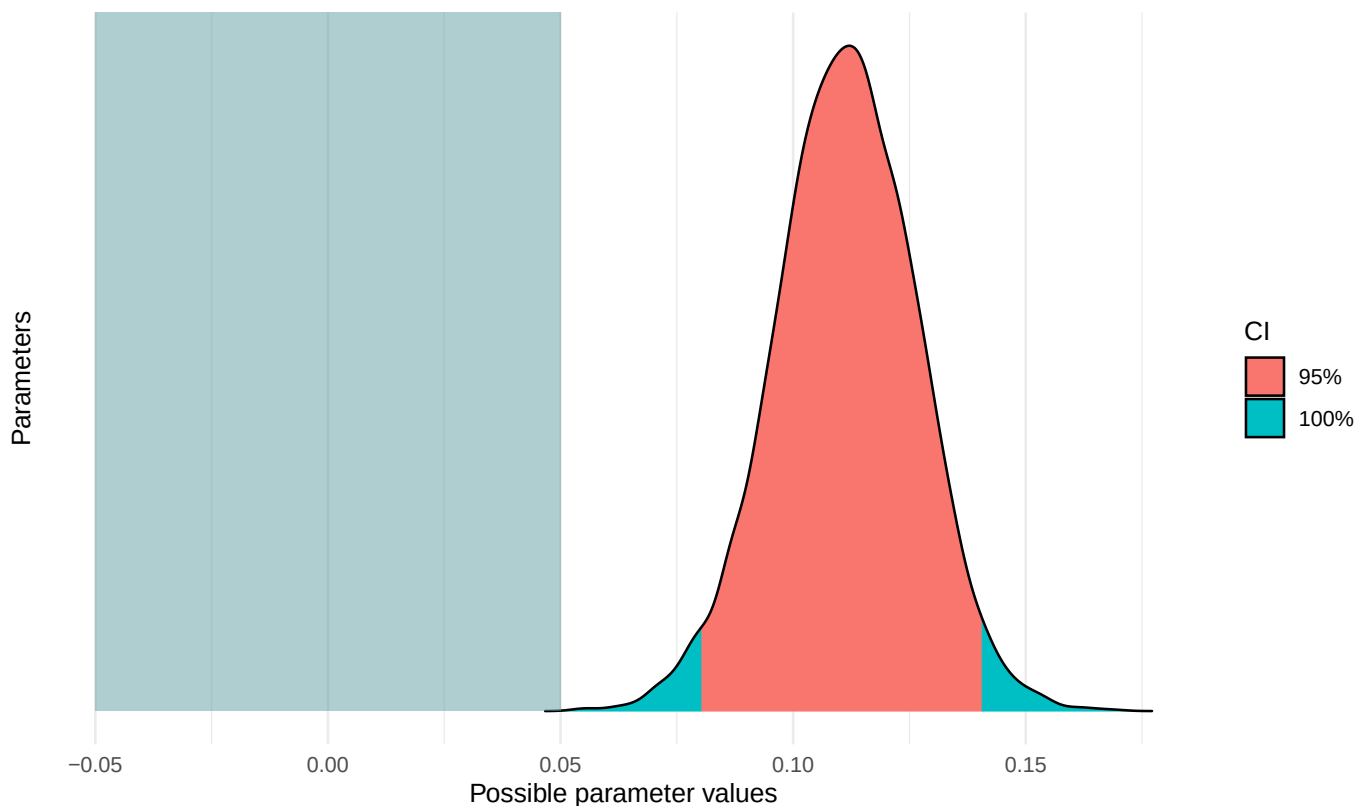
3.4.1 Interpretation

The output shows:

- **% in ROPE:** Proportion of 95% HDI inside ROPE
- Interpretation automatically provided

3.4.2 Visualizing with bayestestR

ROPE Analysis using `bayestestR`



3.4.3 Equivalence Test

The `equivalence_test()` function provides an integrated view:

```
# Test for Practical Equivalence
```

```
ROPE: [-0.05 0.05]
```

Parameter	H0	inside ROPE	95% HDI
Intercept	Rejected	0.00 %	[5.90, 6.05]
conditionB	Rejected	0.00 %	[0.08, 0.14]

This shows for each parameter:

- **% in ROPE:** How much of the posterior falls in ROPE
- **Decision:** Accepted/Rejected/Undecided

i Comparison: Manual vs bayestestR

Both approaches should give the same results:

```
# A tibble: 2 x 3
  Source          ` % in ROPE ` Decision
  <chr>              <dbl>   <chr>
1 Manual Calculation      0 Rejected
2 bayestestR Package      0 Rejected
```

What Just Happened?

Both approaches give identical results — bayestestR simply automates the manual calculations we did earlier. The benefit: - Less code to write - Automatic formatting - Built-in visualization - Consistent interface across analyses

When to Use Manual vs Package

Use manual calculation when:

- You want to understand the mechanics
- You need custom ROPE boundaries for different parameters
- You're teaching/explaining the concept

Use bayestestR when:

- You want quick, standardized results
- You're analyzing many models
- You want built-in visualization

i Prior Sensitivity Analysis for ROPE Decisions

When Priors Matter

From Kruschke (2015, Section 12.2.1): ROPE conclusions can change with different priors.

Always check sensitivity, especially when:

- Sample size is small ($n < 30$ per group)
- HDI barely touches ROPE boundary
- You're accepting H (claiming negligible effect)

3.4.4 Demonstration: Same Data, Different Priors, Different ROPE Conclusions

Fitting models with different priors...

(This may take 1-2 minutes)

	Prior	CI	ROPE_low	ROPE_high	ROPE_Percentage
1	Weakly Informative $N(0, 0.5)$	0.95	-0.05	0.05	1
2	Skeptical $N(0, 0.2)$	0.95	-0.05	0.05	1
3	Diffuse $N(0, 1)$	0.95	-0.05	0.05	1

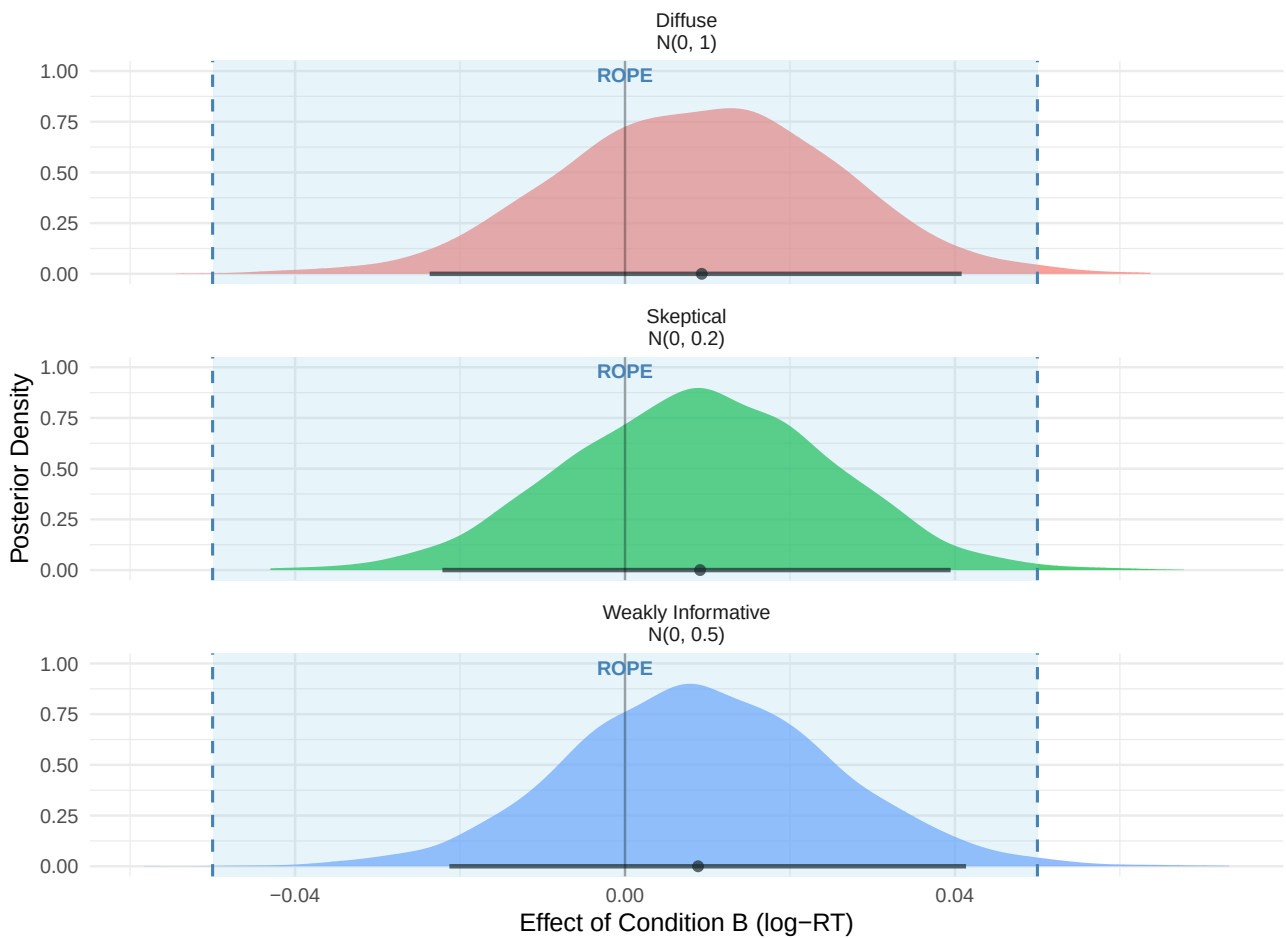
Decision

- 1 Undecided
- 2 Undecided
- 3 Undecided

3.4.5 Visualize Prior Sensitivity

ROPE Sensitivity to Prior Choice

Same data (N=15 subjects), different priors -> How much do conclusions change?



3.4.6 When ROPE Conclusions Hold vs. Change

ROPE Conclusions Hold When:

Large sample size ($n > 30$ per group) - Data overwhelms prior - Posterior similar regardless of prior choice

Effect clearly outside or inside ROPE - HDI far from ROPE boundaries - All reasonable priors give same conclusion

Narrow posterior - HDI width $\ll$ ROPE width - High precision reduces prior influence

Example of robustness:

Data: N=50 subjects, effect = 0.12 log-units

ROPE: [-0.05, 0.05]

Result: 0% in ROPE for all three priors

→ Conclusion holds

ROPE Conclusions Are Sensitive When:

Small sample size ($n < 20$ per group) - Prior has substantial influence - Different priors can change conclusion

HDI barely touches ROPE boundary - Effect 0.05 (right at threshold) - Small prior differences flip decision

Wide posterior - HDI width $\approx$ ROPE width - Uncertainty dominates

Example of sensitivity:

Data: N=15 subjects, effect 0.04 log-units

ROPE: [-0.05, 0.05]

Result:

- Skeptical prior: 100% in ROPE (accept H)
- Diffuse prior: 60% in ROPE (undecided)

→ Conclusion sensitive!

3.4.7 What To Do If Conclusions Are Sensitive

Option 1: Report Multiple Analyses (Recommended)

Report all results from different prior specifications:

- **Weakly informative prior:** [result]
- **Skeptical prior:** [result]
- **Diffuse prior:** [result]
- **Conclusion:** Robust/Sensitive to prior choice

```
# Example code structure
priors_to_test <- list(
  weak = c(prior(normal(0, 0.5), class = b)),
  skeptical = c(prior(normal(0, 0.2), class = b)),
  diffuse = c(prior(normal(0, 1), class = b))
)

# Fit models with each prior and compare ROPE decisions
```

Option 2: Justify Prior More Carefully

If conclusions are sensitive, invest more effort in prior justification:

- Pilot data
- Expert elicitation
- Previous literature
- Prior predictive checks (Module 02)

Report: “We used prior [X] because [justification]. Given potential sensitivity with small N, we verified conclusions hold with more diffuse prior.”

Option 3: Collect More Data

If: - Conclusions sensitive AND - You cannot justify prior AND

- Decision matters
- Collect more data before making decision

Power analysis for ROPE:

```
# Rule of thumb: HDI width inversely proportional to sqrt(n)
current_n <- 15
current_hdi_width <- 0.08
desired_hdi_width <- 0.04 # Want to halve width

target_n <- current_n * (current_hdi_width / desired_hdi_width)^2

tibble(
  Measure = c("Current N", "Current HDI width", "Desired HDI width", "Target N needed"),
  Value = c(current_n, current_hdi_width, desired_hdi_width, round(target_n))
)
```

Option 4: Be Transparent About Uncertainty

If data collection isn't possible:

"Given our sample size (N=15), ROPE conclusions were sensitive to prior specification. With weakly informative priors, X% of posterior fell in ROPE, suggesting [interpretation]. With more skeptical priors, Y% fell in ROPE, suggesting [alternative interpretation]. We recommend treating this finding as preliminary pending replication with larger sample."

3.4.8 Worked Example: Prior Sensitivity Reporting

```
# A tibble: 3 x 4
  Prior                                Estimate `95% HDI` `% in ROPE`
  <chr>                                <dbl> <chr>          <dbl>
1 1. Weakly informative: N(0, 0.5)      NA [NA, NA]          1
2 2. Skeptical: N(0, 0.2)              NA [NA, NA]          1
3 3. Diffuse: N(0, 1)                 NA [NA, NA]          1
```

Conclusion: ROPE decision holds across prior specifications. All three priors yield: Undecided

Main Point

Prior sensitivity is not a bug—it's a feature!

It tells you when your data are: - **Strong enough** to overcome prior beliefs → Confident conclusions - **Too weak** to overcome prior beliefs (sensitive) → Need more data or transparency

NHST doesn't have this diagnostic—it can give you " $p < 0.05$ " even with barely any data, hiding the fact that different assumptions would give different answers.

Bayesian analysis makes sensitivity **explicit and quantifiable**.

4 Understanding Why ROPE Works: Decision Theory

From Practice to Theory

Now that you’ve seen ROPE in action, let’s understand **why** it works. This section explains the decision-theoretic foundations. While not required for using ROPE, it helps you understand what ROPE boundaries mean and when to adjust them.

4.1 The Decision-Theoretic Framework

4.1.1 Making Decisions Under Uncertainty

When we analyze data, we’re ultimately making **decisions**:

- Should we claim an effect exists and is meaningful?
- Should we claim it’s negligible?
- Should we collect more data?

Each decision has **consequences** (utilities or losses):

- **Claiming meaningful effect when it’s negligible**: False positive (wasted resources, misleading literature)
- **Claiming negligible effect when it’s meaningful**: False negative (missed opportunity, wrong theory)
- **Remaining undecided**: Cost of additional data collection

This is **Bayesian decision analysis** (Gelman et al., 2013; see [Stan User’s Guide](#)):

Theoretical Framework (Four Components):

1. **Define outcomes and decisions**
 - Outcomes: Possible values of the effect ()
 - Decisions: {Accept H_0 (negligible), Accept H_1 (meaningful), Remain undecided}
2. **Define probability distribution**
 - Our posterior distribution $p(\theta \mid \text{Data})$
 - Quantifies uncertainty about the effect size
3. **Define utility (or loss) function**
 - $U(\theta, \text{decision})$: What’s the cost/benefit of each decision for each possible true effect?
 - This is where **domain knowledge** enters!
4. **Choose optimal decision**
 - Maximize expected utility (minimize expected loss)
 - $d^* = \arg \max_d E[U(\theta, d) \mid \text{Data}]$
 - (Translation: “Choose decision d that maximizes expected utility, averaging over posterior uncertainty in θ ”)

4.1.2 ROPE as Decision Analysis

ROPE implements this framework!

When you set ROPE boundaries at $[-0.05, +0.05]$, you’re defining a **loss function**:

`Loss( , decision):`

If decide "meaningful effect":

- Low loss if $|| > 0.05$ (correct decision)
- HIGH LOSS if $|| < 0.05$ (false positive)

If decide "negligible effect":

- Low loss if $|| < 0.05$ (correct decision)
- HIGH LOSS if $|| > 0.05$ (false negative)

The **ROPE decision rule** minimizes expected loss:

- If 95% HDI excludes ROPE → High expected utility for “meaningful effect”
- If 95% HDI inside ROPE → High expected utility for “negligible effect”
- If overlapping → Insufficient evidence, collecting more data may have higher utility

! Why This Matters

ROPE is not arbitrary! It's the Bayesian optimal decision given:

1. Your posterior beliefs (from the model)
2. Your utility function (encoded in ROPE boundaries)
3. Your decision threshold (e.g., 95% credibility)

Changing ROPE boundaries = changing your utility function = changing what you consider “meaningful enough to matter”.

5 Effect Estimation with emmeans

i Moving Beyond Single Comparisons

So far we've tested practical significance for a single effect (Condition B vs A). But what if you have **three or more groups**? You need: - All pairwise comparisons (A vs B, A vs C, B vs C) - ROPE analysis for each comparison - Adjustment for multiple comparisons (optional)

This is where **emmeans** and **marginaleffects** come in.

5.1 Why emmeans?

When you have **factorial designs**, you often want to: - Compare all pairwise combinations - Estimate marginal means (averaged over random effects) - Get automatic adjustment for multiple comparisons - Use familiar syntax from frequentist stats

emmeans (estimated marginal means) provides all of this and works directly with brms!

5.2 Example: Three-Condition Design

Let's extend our example to three conditions:

```
# A tibble: 3 x 5
  condition mean_rt sd_rt mean_log_rt sd_log_rt
<fct>      <dbl> <dbl>      <dbl>      <dbl>
1 A          411.  78.8         6.00       0.193
2 B          448.  99.7         6.08       0.221
```

3 C 483. 102. 6.16 0.218

5.3 Estimated Marginal Means

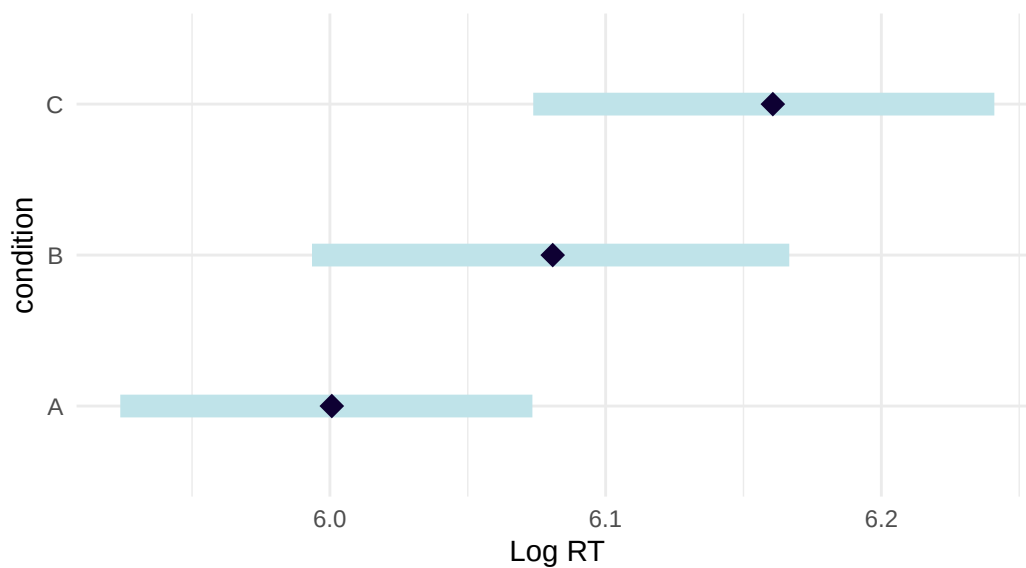
```
condition emmean lower.HPD upper.HPD
A          6.00    5.92    6.07
B          6.08    5.99    6.17
C          6.16    6.07    6.24
```

Point estimate displayed: median

HPD interval probability: 0.95

Estimated Marginal Means

with 95% credible intervals



i What are “Estimated Marginal Means”?

EMMs are model-predicted means that: - Average over random effects (subjects, items) - Provide population-level estimates - Include full Bayesian uncertainty (not just point estimates!)

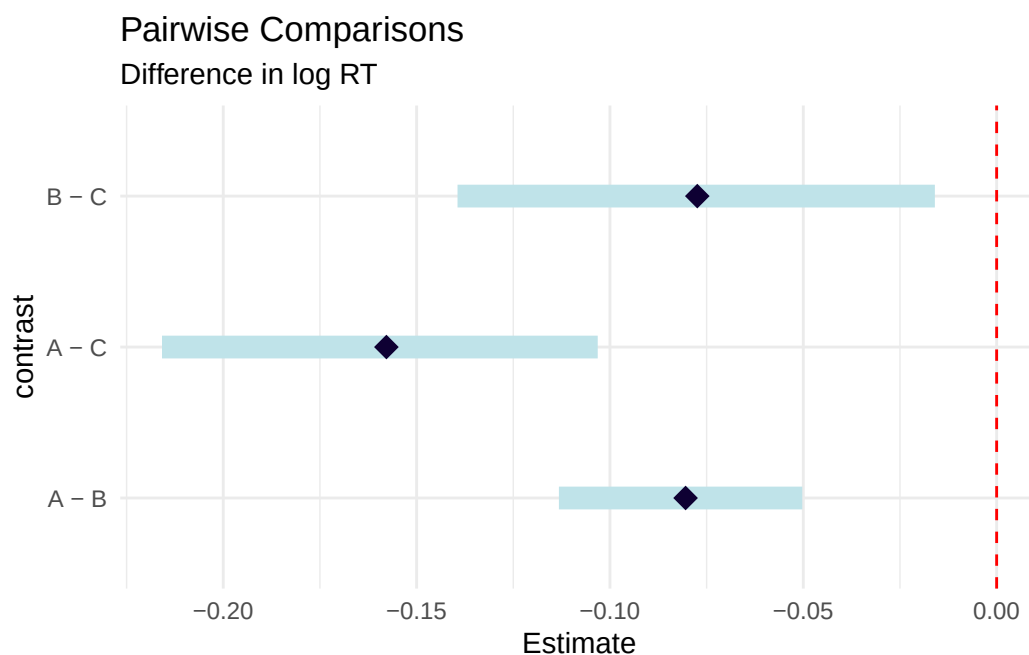
Think of them as “What would we expect for a typical new subject/item?”

5.4 Pairwise Comparisons

```
contrast estimate lower.HPD upper.HPD
A - B      -0.0804   -0.113   -0.0503
A - C      -0.1578   -0.216   -0.1032
B - C      -0.0774   -0.139   -0.0160
```

Point estimate displayed: median

HPD interval probability: 0.95



5.5 ROPE Analysis on Pairwise Comparisons

Now combine emmeans with ROPE to test practical significance of each comparison:

```
↓ 1
  Decision: MEANINGFUL DECREASE

↓ 2
  Decision: MEANINGFUL DECREASE

? 3
  Decision: UNDECIDED
```

5.6 Custom Contrasts

emmeans allows custom contrasts beyond pairwise comparisons:

```
contrast estimate lower.HPD upper.HPD
BC_vs_A    0.1192    0.0857    0.153
C_vs_B     0.0774    0.0160    0.139
```

```
Point estimate displayed: median
HPD interval probability: 0.95
```

```
# A tibble: 2 x 2
  Contrast Decision
  <chr>    <chr>
1 1      Meaningful positive effect
2 2      Undecided
```

Summary: emmeans

You should now understand: - Estimated marginal means (EMMs) are population-level predictions - `pairs()` gives all pairwise comparisons automatically - Extract posterior samples with `as.mcmc()` to integrate with ROPE - Works directly with brms for full Bayesian inference

When to Use emmeans

Perfect for: - Factorial designs (2×2 , 2×3 , etc.) - All pairwise comparisons automatically - Familiar syntax from lsmeans/emmeans in frequentist stats - Integration with ROPE for practical significance
Not ideal for: - Simple two-group comparisons (just use brms coefficients) - Complex non-linear predictions (use `marginalEffects` instead) - Interactions with continuous predictors (`marginalEffects` better)

6 Effect Estimation with `marginalEffects`

Modern Alternative to emmeans

While emmeans is excellent for factorial designs, **`marginalEffects`** offers: - Unified syntax across ALL model types (not just brms) - More flexible predictions and comparisons - Better support for continuous predictors and interactions - Modern tidyverse-compatible workflow
Let's see how it compares.

6.1 Why `marginalEffects`?

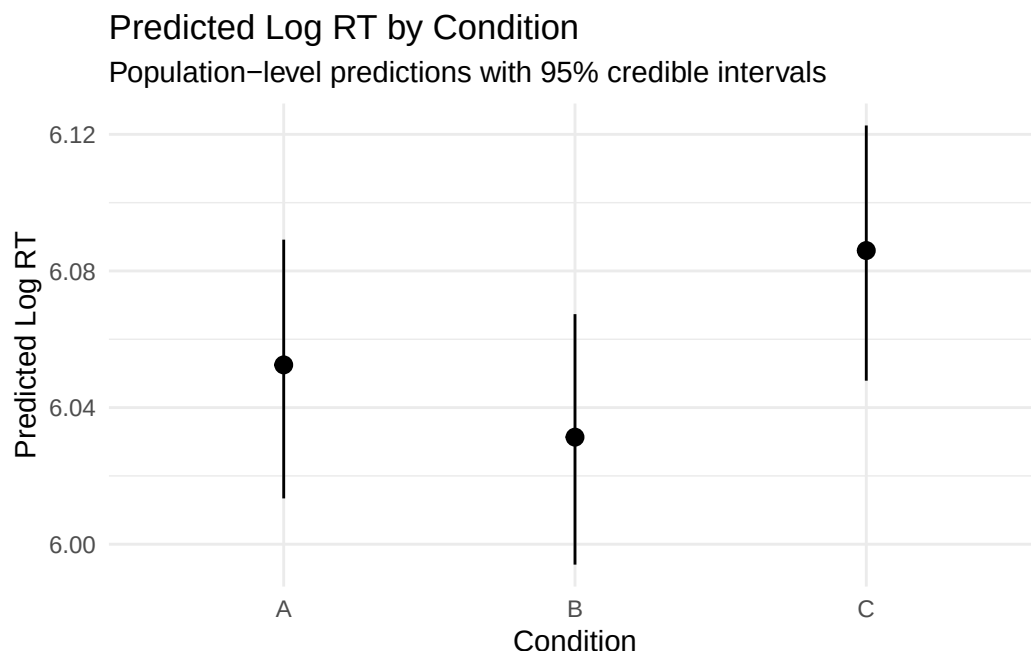
`marginalEffects` provides a modern, unified interface for: - Predictions at specific values - Comparisons (differences, ratios, etc.) - Slopes (derivatives) for continuous predictors - Custom hypotheses with flexible syntax

It works with brms, rstanarm, glm, lme4, and many other models!

6.2 Predictions: What RT Do We Expect?

```
condition Estimate 2.5 % 97.5 %
      A      6.05  6.01  6.09
      B      6.03  5.99  6.07
      C      6.09  6.05  6.12
```

Type: response



6.3 Comparisons: How Much Do Conditions Differ?

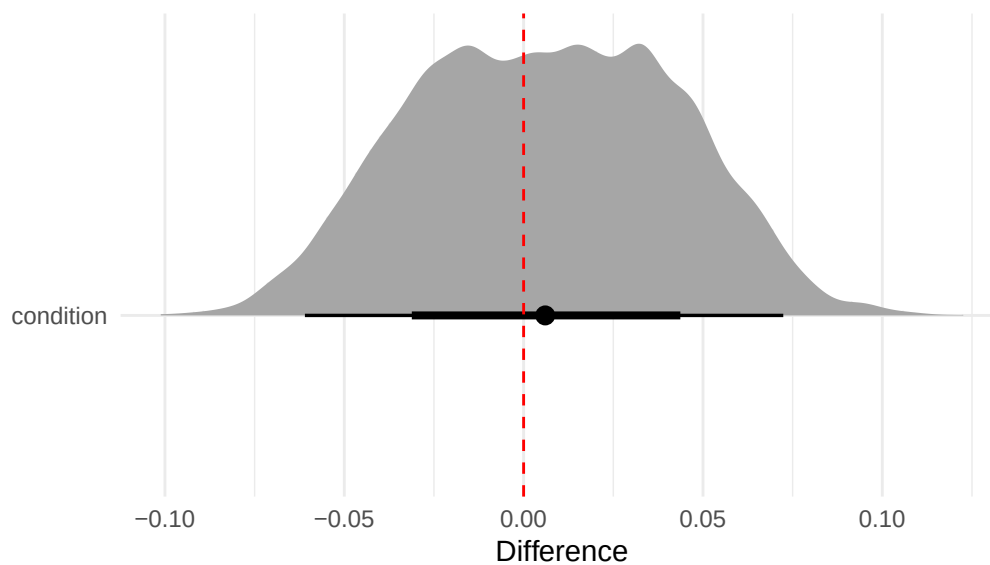
rowid	term	contrast	estimate
Min. :1.00	Length:6	Length:6	Min. :-0.020846
1st Qu.:1.25	Class :character	Class :character	1st Qu.: -0.020846
Median :2.00	Mode :character	Mode :character	Median : 0.006278
Mean :2.00			Mean : 0.006278
3rd Qu.:2.75			3rd Qu.: 0.033402
Max. :3.00			Max. : 0.033402

conf.low	conf.high	item	log_rt	subject
Min. :-0.06844	Min. :0.02615	1 :6	Min. :6.079	1 :6
1st Qu.: -0.06844	1st Qu.:0.02615	2 :0	1st Qu.:6.079	2 :0
Median :-0.04145	Median :0.05307	3 :0	Median :6.079	3 :0
Mean :-0.04145	Mean :0.05307	4 :0	Mean :6.079	4 :0
3rd Qu.: -0.01446	3rd Qu.:0.07999	5 :0	3rd Qu.:6.079	5 :0
Max. :-0.01446	Max. :0.07999	6 :0	Max. :6.079	6 :0
		(Other):0		(Other):0

condition	predicted_lo	predicted_hi	predicted	tmp_idx
A:2	Min. :6.053	Min. :6.031	Min. : NA	Min. :1.00
B:2	1st Qu.:6.053	1st Qu.:6.031	1st Qu.: NA	1st Qu.:1.25
C:2	Median :6.053	Median :6.059	Median : NA	Median :2.00
	Mean :6.053	Mean :6.059	Mean :NaN	Mean :2.00
	3rd Qu.:6.053	3rd Qu.:6.086	3rd Qu.: NA	3rd Qu.:2.75
	Max. :6.053	Max. :6.086	Max. : NA	Max. :3.00
			NA's :6	

Pairwise Comparisons

Posterior distributions of differences in log RT



6.4 Reference Grid: Custom Comparisons

```

Estimate      2.5 % 97.5 %
-0.0208 -0.06844 0.0261
 0.1046  0.05853 0.1512
 0.0186 -0.02805 0.0639
 0.1531  0.10796 0.1988
 0.0571  0.01368 0.1010
--- 1790 rows omitted. See ?print.marginaleffects ---
 0.0480  0.00257 0.0939
 0.2671  0.22047 0.3140
 0.1750  0.13032 0.2200
 0.0956  0.04961 0.1422
 0.1615  0.11506 0.2089
Term: condition
Type: response
Comparison: B - A

# A tibble: 5 x 2
  Measure      Value
  <chr>        <chr>
1 Estimate      0.044
2 95% HDI      [0.003, 0.085]
3 % Below ROPE 0.0%
4 % In ROPE    61.1%
5 % Above ROPE 38.9%

```

6.4.1 Decision

```
[1] " Undecided - need more data"
```

6.5 Hypotheses: Flexible Hypothesis Testing

`marginaleffects` doesn't support `hypotheses()` for Bayesian models, but we can test hypotheses using comparisons:

Table 5: All Pairwise Comparisons with Posterior Statistics

Comparison	Estimate	Std Error	95% CI		P(diff > 0)	P(diff > 0.05)
			Lower	Upper		
condition	0.1193	0.1178	-0.0496	0.3799	0.880	0.701

6.5.1 Interpretation Guide

- **Estimate:** Mean difference in log-RT between conditions
- **95% CI:** Credible interval for the difference
- **P(diff > 0):** Probability first condition has higher RT than second
- **P(|diff| > 0.05):** Probability difference exceeds ± 0.05 ROPE threshold

6.5.2 Directional Probabilities

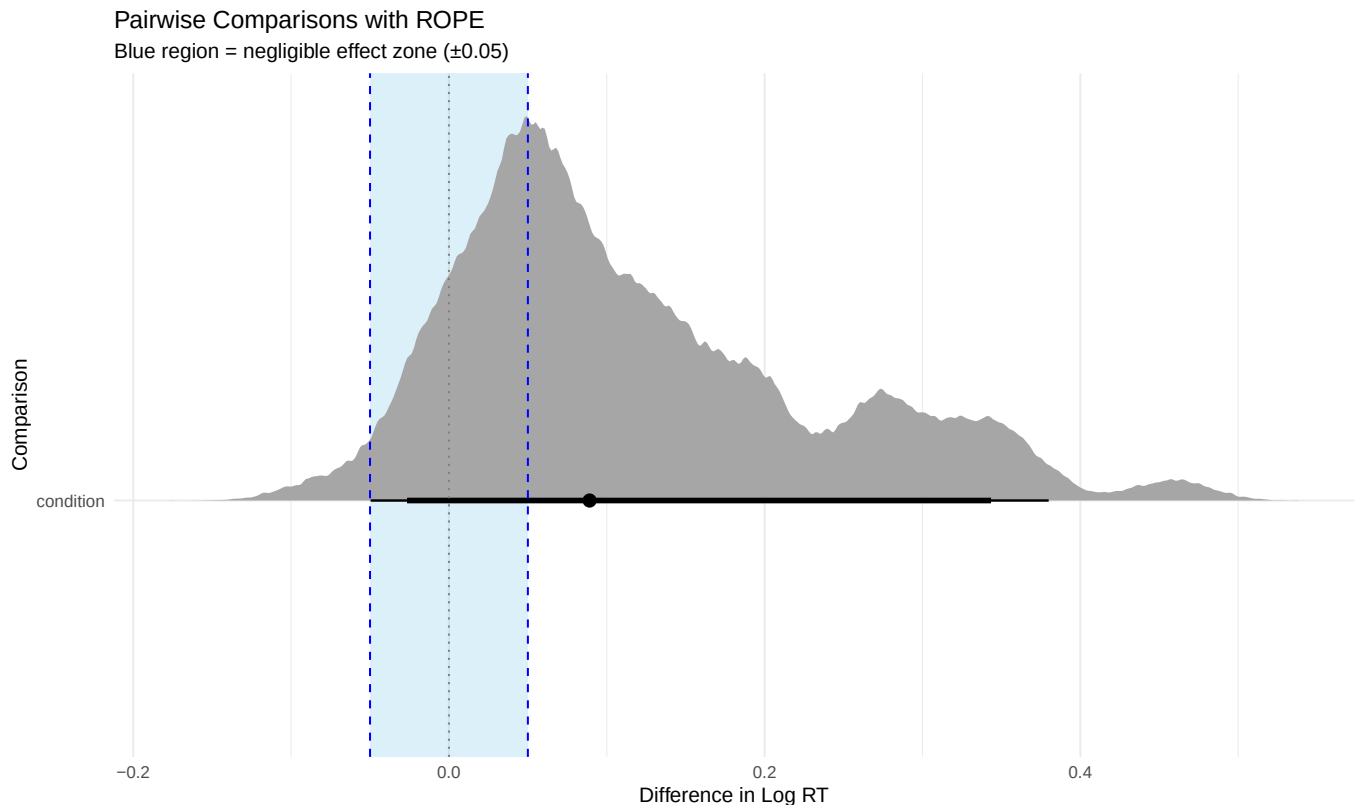
```
# A tibble: 3 x 2
  Comparison Probability
  <chr>             <dbl>
1 P(B > A)          NaN
2 P(C > A)          NaN
3 P(C > B)          NaN
```

6.6 ROPE Analysis with `marginaleffects`

Combine `marginaleffects` with ROPE:

MARGINALEFFECTS + ROPE ANALYSIS

```
condition
Estimate: 0.119 [-0.050, 0.380]
ROPE: 2.5% | In: 29.9% | Above: 67.6%
Decision: UNDECIDED
```



i Summary: `marginalEffects`

You should now understand: - `predictions()` computes model predictions at specific values - `comparisons()` computes differences between conditions - `posterior_draws()` extracts MCMC samples for ROPE analysis - Works with `brms`, `rstanarm`, `glm`, `lme4`, and many other packages

💡 When to Use `marginalEffects`

Perfect for: - Any type of model (GLM, multilevel, GAM, etc.) - Predictions at specific covariate values - Non-linear transformations (odds ratios, percentages, etc.) - Custom hypotheses with complex logic - Modern, consistent syntax across models

Advantages over `emmeans`: - More flexible predictions - Better for continuous predictors - Unified interface across packages - Direct ROPE integration via `posterior_draws()`

Use `emmeans` if: - You want traditional EMM workflow - All pairwise comparisons with adjustment - Familiar with `lsmeans/emmeans` syntax

7 ROPE Analysis Workflow

Bringing It All Together

You've now learned: - ROPE: Theory and implementation - emmeans: Factorial design comparisons
- marginalesffects: Flexible predictions - tidybayes: Visualization
Now let's integrate everything into a complete analysis workflow.

Now that we've covered all the tools (ROPE, emmeans, marginalesffects, tidybayes), let's see how to put them together in a complete practical significance analysis.

7.1 Step-by-Step ROPE Analysis

```
# A tibble: 2 x 2
  Measure      Value
  <chr>        <chr>
1 Posterior mean 0.111
2 95% HDI       [0.084, 0.142]
```

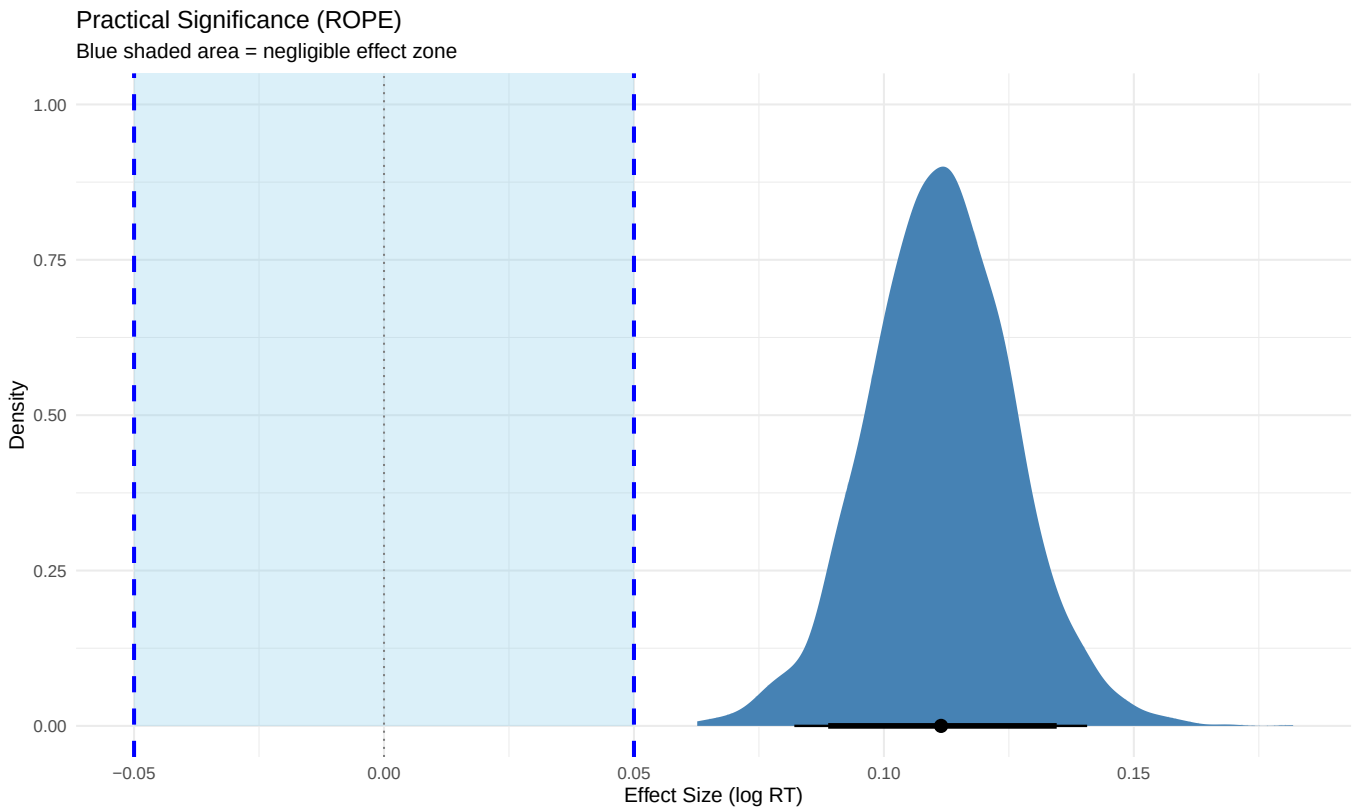
7.1.1 Step 2: Test Practical Significance (ROPE)

```
# A tibble: 3 x 2
  Measure      Value
  <chr>        <chr>
1 ROPE        [-0.05, +0.05] (5% RT difference)
2 % in ROPE 0.0%
3 Decision    Effect is practically meaningful (HDI > ROPE)
```

7.1.2 Step 3: Integrated Conclusion

```
STRONG CONCLUSION: Effect is meaningful
→ Condition B differs from A
→ The difference is large enough to matter
```

7.2 Visualizing the Complete Picture



7.3 Complete Reporting: APA-Style Templates

! What to Always Report

From Kruschke (2015, p. 338):

“Reporting the limits of an HDI region is more informative than reporting the declaration of a reject/accept decision. By reporting the HDI and other summary information about the posterior, different readers can apply different ROPEs to decide for themselves whether a parameter is practically equivalent to a null value.”

Complete reporting includes: 1. Full posterior summary (not just inside/outside ROPE) 2. ROPE boundaries with justification 3. Effect size on multiple scales 4. Model diagnostics 5. Sensitivity checks

7.3.1 Methods Section Template

Use this template for your Methods section when reporting ROPE analysis:

Always include in Methods: - [] Prior specification with rationale - [] ROPE boundaries with justification - [] When boundaries were set (a priori) - [] How boundaries relate to original scale - [] Sample size (subjects, items, observations) - [] Software versions

7.3.2 Results Section Template

Use this template for your Results section:

Always include in Results: - [] Point estimate with HDI (on model scale) - [] Posterior SD or uncertainty measure - [] % of posterior in/outside ROPE - [] Decision statement (reject/accept/undecided H) - [] Effect size on original scale with interpretation - [] Absolute magnitudes (e.g., milliseconds) where relevant - [] Robustness checks (prior sensitivity, model comparison)

7.3.3 Complete Example: Full Reporting

Here's a complete example combining Methods and Results:

7.3.4 What NOT to Report (Common Mistakes)

Too brief (incomplete):

"The effect was meaningful according to ROPE ($p < .05$ equivalent)."

Problem: No effect size, no HDI, no uncertainty quantification

Binary decision only:

"The 95% HDI excluded the ROPE, so we reject the null hypothesis."

Problem: Doesn't report actual posterior, prevents readers from applying own ROPE

Missing justification:

"Using ROPE = $[-0.05, 0.05]$, the effect was meaningful."

Problem: No justification for ROPE boundaries, no when specified

Complete and correct:

"Effect: $\beta = 0.12$ (95% HDI: $[0.09, 0.15]$), corresponding to 12.7% RT increase. With ROPE = ± 0.05 (justified by perceptual threshold from pilot, $N=20$), 100% of posterior mass fell outside ROPE, indicating practical meaningfulness. Insensitive to prior specification (tested with $N(0, 0.2)$ and $N(0, 1)$ priors)."

7.3.5 Reporting When Undecided

If your ROPE analysis is undecided (HDI overlaps ROPE), report honestly:

7.4 Practical Reporting Template

When writing up your results, focus on practical significance:

8 When to Use What: Decision Framework

i What You've Learned

By now you can: - Conduct ROPE analysis to test practical significance - Use `emmeans` for factorial designs - Use `marginalEffects` for flexible comparisons - Combine ROPE with any comparison tool - Visualize results with `tidybayes` - Report practical significance in publications

💡 Quick Decision Tree

Question 1: What do you want to test?

- “Is the effect big enough to matter?” → Use ROPE
- “Which hypothesis is better supported?” → See Module 07 (Bayes Factors)
- “Which model fits better?” → Use LOO (Module 03)

Question 2: What’s your research goal?

- Practical decision-making → ROPE
- Theory testing (comparing hypotheses) → Module 07
- Publishing/reporting → ROPE + model comparison (LOO)

Question 3: What have you learned so far in this workshop?

- Module 03: Model comparison with LOO → For selecting between competing models
- Module 04: Prior sensitivity → For checking robustness
- Module 05: Prior predictive checks → For setting reasonable priors
- **Module 06 (now):** ROPE → For testing practical significance
- **Module 07 (next):** Bayes Factors → For hypothesis comparison

8.1 Visual Guide: Choosing Your Tools

Your Research Question

- "Is effect meaningful?" → ROPE
- "Compare 3+ groups?"
 - Factorial design → emmeans + ROPE
 - Custom predictions → marginaleffects + ROPE
- "Which hypothesis better?" → Module 07 (Bayes Factors)
- "Which model structure?" → Module 05 (LOO)

8.2 Detailed Comparison

Table 6: Choosing the Right Tool for Your Research Question

Tool	Research Question	Output	Use When
ROPE	Is effect negligible?	% in ROPE, decision rule	Test practical significance
emmeans	What are all pairwise differences?	All pairwise comparisons, EMMs	Factorial designs
marginaleffects	What’s the effect at X?	Predictions, custom contrasts	Flexible predictions/comparisons
LOO (Module 03)	Which model predicts better?	elpd difference, model weights	Comparing model structures
Bayes Factors (Module 07)	Which hypothesis is better?	Evidence ratio, posterior prob	Comparing specific hypotheses

8.3 Integration with Previous Modules

8.3.1 Module 03: Model Comparison with LOO

When to use LOO vs ROPE:

- **LOO:** Which model structure is better? (e.g., with/without random slopes)
- **ROPE:** Within a model, is the effect practically meaningful?

```
# Example combining both:  
# 1. Use LOO to select best model structure  
loo_compare(model_simple, model_complex)  
  
# 2. Then test practical significance in the winning model  
rope(model_complex, ci = 0.95, range = c(-0.05, 0.05))
```

8.3.2 Module 04: Prior Sensitivity

Note: ROPE conclusions can be sensitive to priors for small samples!

```
# Check: Does conclusion change with different priors?  
model_weak_prior <- brm(..., prior = normal(0, 10))  
model_strong_prior <- brm(..., prior = normal(0, 1))  
  
rope(model_weak_prior, range = c(-0.05, 0.05)) # Compare decisions  
rope(model_strong_prior, range = c(-0.05, 0.05))
```

8.3.3 Module 05: Prior Predictive Checks

Workflow integration:

1. **Before fitting:** Use prior predictive checks to check priors are reasonable
2. **After fitting:** Use ROPE for practical significance
3. **Robustness:** Check that conclusions hold under alternative priors (Module 04)

8.3.4 Module 07: Bayes Factors (Next)

For hypothesis comparison and evidence ratios, see Module 07 on Bayes Factors

9 Common Pitfalls and How to Avoid Them

Pitfall #1: ROPE Too Narrow or Too Wide

The Problem:

```
# Too narrow: Almost nothing is "negligible"  
rope(model, range = c(-0.01, 0.01))  
  
# Too wide: Everything is "negligible"  
rope(model, range = c(-1, 1))
```

The Solution:

```
# Base ROPE on smallest effect size of practical interest (SESOI)
# Example: For Cohen's d, often use ±0.1 or ±0.2
rope(model, range = c(-0.1, 0.1)) # SESOI = 0.1
```

Why: ROPE should reflect domain knowledge, not statistical convenience.

⚠ Pitfall #2: Post-Hoc ROPE Boundaries

The Problem:

```
# WRONG: Looking at posterior first, then choosing ROPE
summary(model) # Hmm, effect is around 0.08...
rope_result <- rope(model, range = c(-0.07, 0.07)) # Chosen to exclude effect!
```

The Solution:

- Set ROPE boundaries **before** looking at results
- Base on domain knowledge, not convenience
- Document your rationale

Examples of good justification:

- “Previous meta-analysis showed minimum detectable effect = 0.5 SD”
- “Cost-benefit analysis requires >10% improvement to justify intervention”
- “Perception threshold for RT differences is ~50ms”

⚠ Pitfall #3: Ignoring ROPE Overlap

The Problem:

95% HDI overlaps ROPE, but you want a clear answer!

```
# 95% HDI: [-0.02, 0.12], ROPE: [-0.05, 0.05]
# Decision: UNDECIDED
```

Tempting (wrong) responses:

- “Let’s just call it meaningful” (ignores uncertainty)
- “Let’s widen ROPE until it fits” (post-hoc manipulation)
- “Let’s use a different test instead” (selective reporting)

Appropriate responses:

- Report the uncertainty honestly
- Collect more data if possible
- Use decision theory (e.g., “accept uncertainty and proceed”)
- Consider using 89% or 90% HDI (less conservative)

9.1 Avoiding Pitfalls: Checklist

Before running analyses:

- ☐ Priors are **weakly informative** (not flat)?
- ☐ ROPE boundaries defined **before** seeing results?
- ☐ ROPE boundaries **justified** with domain knowledge?

- ☐ Hypotheses based on **theory**, not data exploration?

When interpreting results:

- ☐ Report **ROPE decision** and effect sizes?
- ☐ Report **uncertainty** (don't hide ROPE overlaps)?
- ☐ Check **prior sensitivity** (Module 04)?
- ☐ Show **posterior distributions visually**?
- ☐ Interpret on the **original scale** when possible?

10 Summary and Best Practices

10.1 Summary

1. **ROPE distinguishes practical from statistical significance**
 - Must be defined a priori based on domain knowledge
 - Provides decision framework for inference
2. **emmeans simplifies factorial design analysis**
 - Automatic pairwise comparisons
 - Familiar syntax for traditional stats users
 - Seamless integration with ROPE
3. **marginaleffects provides flexible predictions**
 - Unified interface across model types
 - Custom contrasts and transformations
 - Direct posterior access for ROPE
4. **Combine all three for better inference**
 - ROPE: “Is it meaningful?”
 - emmeans: “What are all the differences?” (factorial designs)
 - marginaleffects: “What happens at specific values?” (flexible)
5. **For hypothesis comparison, see Module 07**
 - Bayes Factors for comparing specific hypotheses
 - Evidence ratios using Savage-Dickey method

10.2 Complete Bayesian Inference Workflow (All Modules)

```
# 1. Fit model
model <- brm(y ~ condition + (1 + condition | subject), data = data, ...)

# 2. Check model (Module 03-04)
pp_check(model)
prior_sensitivity(model)

# 3. Test practical significance with ROPE
rope(model, range = c(-0.05, 0.05))

# 4a. If factorial design: Use emmeans
emm <- emmeans(model, ~ condition)
pairs(emm) # All pairwise comparisons
```

```
# 4b. If custom predictions: Use marginalesffects
predictions(model, newdata = datagrid(condition = c("A", "B")))
comparisons(model, variables = "condition")

# 5. Combine with ROPE for each comparison
# (see examples above)

# 6. Visualize everything
plot(emm) # or plot(predictions(...))
# + Add ROPE boundaries to visualization

# 7. Report clearly
# - Effect size with HDI
# - ROPE decision
# - Interpretation on original scale
```

- Statistical evidence (hypothesis test)
- Practical significance (ROPE)
- Transparent reporting of both

11 Summary: ROPE as Decision Analysis

11.1 The Conceptual Framework

ROPE is not just a hypothesis test — it’s a decision framework under uncertainty.

Every ROPE analysis follows the four-step Bayesian decision analysis framework (Gelman et al., 2013; [Stan User’s Guide](#)):

1. **Define outcomes** → Your posterior distribution $p(\theta | \text{Data})$
2. **Define decisions** → {Accept meaningful effect, Accept negligible effect, Remain undecided}
3. **Define utility function** → Encoded in your ROPE boundaries (domain knowledge!)
4. **Choose optimal decision** → Maximize expected utility (minimize expected loss)

Note: When you set ROPE boundaries at $[-0.05, +0.05]$, you’re implicitly defining a loss function:

- **Inside ROPE:** Effects here have low practical utility (too small to matter)
- **Outside ROPE:** Effects here have high practical utility (worth acting on)
- **ROPE boundary:** Your smallest effect size of interest (SESOI)

The traditional ROPE decision rule (“**95% HDI must exclude/include ROPE**”) approximates the Bayes-optimal decision under: - Approximately symmetric losses for false positives vs. false negatives - Acceptable 5% error rate

For asymmetric losses (e.g., false positives much more costly than false negatives): - Adjust the HDI threshold (use 99% instead of 95%), OR - Compute expected utilities explicitly (as shown in this module)

11.2 Three Tools, One Goal: Practical Significance

This module covered three complementary approaches:

1. **ROPE**: Test if effect is negligible vs. meaningful
 - Decision framework for practical equivalence
 - Based on domain-defined boundaries
 - Handles three outcomes: Accept H , Accept H , Undecided
2. **emmeans**: Compare groups in factorial designs
 - All pairwise comparisons with adjustment
 - Estimated marginal means
 - Integrates with ROPE for each comparison
3. **marginaleffects**: Flexible predictions and contrasts
 - Unified interface for any model type
 - Custom hypotheses and transformations
 - Modern approach to effect estimation

Use **all three together** for better inference: - ROPE quantifies practical significance - emmeans/marginaleffects estimate effects and compare groups - Together they answer: “How big is it?” AND “Does it matter?”

11.3 Connection to Other Modules

Module 07 (Next): Bayes Factors - ROPE asks: “Is this effect meaningful?” - Bayes Factors ask: “Which hypothesis is better supported?” - **Different questions, complementary approaches!**

Module 05: LOO Cross-Validation - LOO: Which model predicts better? - ROPE: Is the effect practically meaningful? - Use LOO for model selection, ROPE for inference

Module 04: Prior Sensitivity - Check that ROPE conclusions hold under different priors - Especially needed for small samples

11.4 The Bottom Line

Before ROPE:

```
summary(model)
# Effect:   = 0.08, 95% CI: [0.05, 0.11]
# Conclusion: "The effect is detectable"
# Reader: "But does it matter?"
```

With ROPE:

```
rope(model, range = c(-0.05, 0.05))
# 95% HDI: [0.05, 0.11] excludes ROPE
# Conclusion: "The effect is practically meaningful"
# Justification: "Effects > 5% RT difference are perceptible"
```

ROPE provides: - Clear decision framework (not just p-values) - Domain knowledge integration (ROPE boundaries) - Three outcomes (accept H , accept H , undecided) - Theoretical foundation (Bayesian decision theory)

Remember: ROPE boundaries must be set a priori based on domain knowledge, not post-hoc to get desired results!

11.5 Workflow Checklist

- ☐ Define hypotheses a priori based on theory
- ☐ Set ROPE boundaries based on domain knowledge (not data)
- ☐ Fit model with appropriate priors
- ☐ Conduct ROPE analysis on effects of interest
- ☐ Visualize posterior distributions with ROPE
- ☐ Report practical significance with context
- ☐ For hypothesis comparison, see Module 07
- ☐ Interpret results in context of theory and application

11.6 Literature and Resources

For references, tutorials, and resources on ROPE analysis, Bayesian modeling with brms, and related topics, see:

[Complete Literature and Resources Guide](#)

11.6.1 Quick Reference

Main papers: - **Kruschke, J. K. (2018).** Rejecting or accepting parameter values in Bayesian estimation. *Advances in Methods and Practices in Psychological Science*, 1(2), 270-280. [The definitive ROPE paper] - **Kruschke, J. K. (2015).** *Doing Bayesian data analysis* (2nd ed.). Academic Press. [Chapters 11-12] - **Makowski, D., Ben-Shachar, M. S., & Lüdtke, D. (2019).** bayestestR: Describing effects and their uncertainty. *Journal of Open Source Software*, 4(40), 1541. - **Lakens, D., Scheel, A. M., & Isager, P. M. (2018).** Equivalence testing for psychological research. *Advances in Methods and Practices in Psychological Science*, 1(2), 259-269.

11.7 Session Info

R version 4.5.2 (2025-10-31)

Platform: x86_64-pc-linux-gnu

Running under: Ubuntu 24.04.3 LTS

Matrix products: default

BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3

LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.26.so; LAPACK version 3.12.0

locale:

```
[1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
[3] LC_TIME=en_US.UTF-8       LC_COLLATE=en_US.UTF-8
[5] LC_MONETARY=en_US.UTF-8   LC_MESSAGES=en_US.UTF-8
[7] LC_PAPER=en_US.UTF-8      LC_NAME=C
[9] LC_ADDRESS=C              LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

time zone: Etc/UTC

tzcode source: system (glibc)

attached base packages:


```
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

other attached packages:

```
[1] knitr_1.50          patchwork_1.3.2      bayestestR_0.17.0
[4] tidybayes_3.0.7     posterior_1.6.1.9000 bayesplot_1.15.0
[7] lubridate_1.9.4     forcats_1.0.1        stringr_1.5.2
[10] dplyr_1.1.4         purrr_1.1.0          readr_2.1.5
[13] tidyr_1.3.2         tibble_3.3.0         ggplot2_4.0.1
[16] tidyverse_2.0.0     brms_2.23.0          Rcpp_1.1.0
```

loaded via a namespace (and not attached):

```
[1] svUnit_1.0.8        tidysselect_1.2.1    farver_2.1.2
[4] loo_2.8.0           S7_0.2.0             fastmap_1.2.0
[7] tensorA_0.36.2.1    digest_0.6.37        timechange_0.3.0
[10] estimability_1.5.1  lifecycle_1.0.4      StanHeaders_2.32.10
[13] processx_3.8.6      magrittr_2.0.4       compiler_4.5.2
[16] rlang_1.1.6         tools_4.5.2          utf8_1.2.6
[19] yaml_2.3.10         data.table_1.17.8    labeling_0.4.3
[22] bridgesampling_1.2-1 pkgbuild_1.4.8        plyr_1.8.9
[25] RColorBrewer_1.1-3  cmdstanr_0.9.0       abind_1.4-8
[28] withr_3.0.2         datawizard_1.3.0     grid_4.5.2
[31] stats4_4.5.2        xtable_1.8-4         inline_0.3.21
[34] ggribes_0.5.7       emmeans_2.0.1        scales_1.4.0
[37] tinytex_0.57        insight_1.4.4        cli_3.6.5
[40] mvtnorm_1.3-3       rmarkdown_2.30       generics_0.1.4
[43] RcppParallel_5.1.11-1 reshape2_1.4.5       tzdb_0.5.0
[46] rstan_2.32.7        HDInterval_0.2.4     parallel_4.5.2
[49] matrixStats_1.5.0   vctrs_0.6.5         Matrix_1.7-4
[52] jsonlite_2.0.0      hms_1.1.4            arrayhelpers_1.1-0
[55] ggdist_3.3.3        see_0.12.0           glue_1.8.0
[58] codetools_0.2-20    ps_1.9.1             distributional_0.5.0
[61] stringi_1.8.7       gtable_0.3.6         QuickJSR_1.8.1
[64] pillar_1.11.1       htmltools_0.5.8.1    Brodningnag_1.2-9
[67] R6_2.6.1            evaluate_1.0.5       lattice_0.22-7
[70] backports_1.5.0     rstantools_2.5.0     coda_0.19-4.1
[73] gridExtra_2.3       nlme_3.1-168         checkmate_2.3.3
[76] xfun_0.55           pkgconfig_2.0.3
```